

Simply Energy: Helping Kiwi businesses decarbonize with cloud data and AI solutions

Tell us your challenge. We're here to help.

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About Simply Energy

With Google Cloud, Simply Energy has consolidated and automated and simplified the data management processes needed by our customers on reducing energy costs and emissions.



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Part of the New Zealand-based Contact Energy group, Simply Energy offers advisory support, solutions, and energy supply arrangements to assist its commercial and industrial customers in achieving net zero carbon emissions.

Industries: Energy, Chemicals & Utilities

Location: New Zealand

Google Cloud results

- Reduces data storage costs by 50% to improve budget management
- Reduces testing time for new products and services by 50%
- Lowers coding time by 40% to enable developers to work directly with the business

Deliver faster mode

Google Cloud (<https://cloud.google.com/>),
Looker (<https://cloud.google.com/looker>)
BigQuery (<https://cloud.google.com/bigquery/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Compute Engine
(<https://cloud.google.com/compute/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Looker (<https://cloud.google.com/looker>)

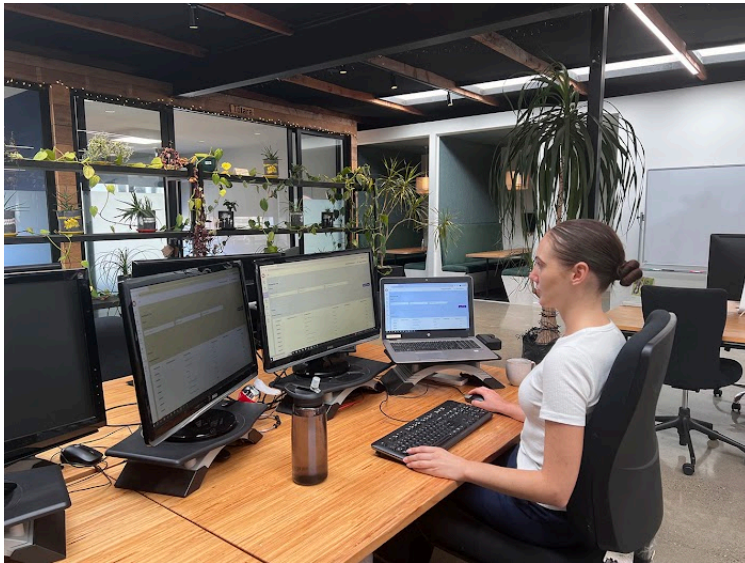
More organizations today are looking for ways to reduce their environmental impact. Helping commercial and industrial customers achieve their 'net zero' targets is the mission of Simply Energy (<https://simplyenergy.co.nz/>), the energy transition-oriented arm of New Zealand electricity, gas and broadband provider Contact Energy.

Capturing data from multiple sources to help drive net zero

With Simply Energy, businesses can access data-driven insights into their power consumption and associated carbon footprint along with advisory services that help bring about meaningful change. Simply ingests data from a large number of sources, in various formats and structures, and at frequencies ranging from seconds to hourly. These sources include New Zealand's grid provider, regulators, electricity meters, billing and other applications, and IoT sensors in customer-facing solutions.

"With the data we collect, we can help our customers quantify the carbon emissions generated from the energy they use," explains Hema Pericherla, Chief Technology Officer at Simply Energy. "We build on that with advisory services that include modeling, generating insights and helping with efficiency measures that reduce emissions."

Using a Google Cloud data architecture and infrastructure, Simply Energy extracts, transforms and loads (ETLs) data from all its sources into a cloud data warehouse for storage, organization, and analysis. This data is then used to support business teams, and data and business analysts on advisory strategies, and generate financial and operational models and reports.



Rebuilding the tech stack to improve customer experiences

After joining forces with Contact Energy in 2021, Simply Energy faced the challenge of integrating two very different technology environments to deliver a seamless experience to Contact's commercial and industrial customer base. It had previously run its business from an on-premises server, while its new parent operated a multi-cloud architecture. "In 2021 we reviewed our technology stack including our cloud strategy and decided to rebuild completely to remove complexity from our data, cloud and platform layers," says Pericherla.

With a tight timeframe to evaluate options for the rebuild, Simply Energy prioritized responsiveness from vendors, cost, technical expertise and competency, and quality of service and support in its procurement process. This process led the business to choose Google Cloud as its preferred cloud provider.

Simply Energy started the build and deployment and completed the exercise with Google Cloud in 2023 with the implementation of data visualization through the Looker business intelligence platform.

"The move to Google Cloud was seamless, and our primary contact there gave us access to technical analysts and specialist teams across multiple continents," says Pericherla.

The business has consolidated to an architecture incorporating [Google Kubernetes Engine](https://cloud.google.com/gke) (<https://cloud.google.com/gke>) to run core applications in a containerized environment and [Compute Engine](https://cloud.google.com/compute) (<https://cloud.google.com/compute>) to power event-driven serverless functions. A [BigQuery](https://cloud.google.com/bigquery) (<https://cloud.google.com/bigquery>) data warehouse manages and cleanses all internal and external data, with a PostgreSQL relational database system used to aggregate the cleansed data. [Cloud SQL](https://cloud.google.com/sql) (<https://cloud.google.com/sql>) provides analytics and [Looker](https://cloud.google.com/looker) (<https://cloud.google.com/looker>) enables visualization and business intelligence.

This architecture has paid off considerably for the business. "We opted for Google Cloud to realize the full potential of our business, and our transformation to date has been overwhelmingly successful," says Pericherla. "For example, scenario-based modeling in the new platform is three times faster than previously, because Google Cloud services can rapidly ingest and analyze our large datasets and handle multiple tasks concurrently.

"Also, Google Cloud machine learning saves time by enabling us to automate and optimize aspects of the data modeling process."

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—Hema Pericherla, Chief
Technology Officer,
Simply Energy

Reducing costs and maintenance loads

In addition, Pericherla says the complexity of the business's technology environment before Google Cloud imposed a high maintenance burden on its technology team, and imposed excessive costs for the size of the business. "Based on the size of

Simply Energy in 2021, the data sources we were using and our activities, these costs were about 50% higher than optimal," she says.

With Google Cloud, the business has slashed its maintenance costs, and achieved considerable savings across its environment. "With BigQuery, our data storage costs are 50 percent lower on a like-for-like basis than the costs we incurred in another cloud service," adds Pericherla.

In addition, Google Cloud tools enable Simply Energy to closely monitor its licenses and overall cloud spend, including cost per query, allowing the organization to optimize coding and query behaviors.

The use of managed cloud services from a single provider has reduced the technology team's maintenance load to a minimum, freeing resources to deliver projects faster than promised to customers.

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Simply Energy*

Delivering meaningful visualizations to meet customer needs

With Looker, Simply Energy can combine data seamlessly into meaningful visualizations in response to ad-hoc or individual customer requests. It has also cut the time to deliver these reports, which could require the collection of energy data from multiple locations, from a month to about two weeks. The business is now working towards providing customers with direct access to charts and maps generated by the business intelligence platform.

With its new Google Cloud environment up and running, Simply Energy has restructured to enable developers to spend up to 40% more of their time working directly with the business to gather requirements. "We have increased our efficiency considerably by establishing these direct relationships rather than asking our developers to

work with intermediaries who translate business requirements," explains Pericherla.

As a business, Simply Energy is always looking to provide new solutions and enhanced features for customers. In the Google Cloud environment, the team has reduced the time to test these features by half, allowing for more runway and quicker go-to-market.



Leveraging AI to decarbonize New Zealand

With Google Cloud, the organization has digitized its business processes to power its customer journeys, experiences, and capability plans, and now it wants to provide a future outlook and cost certainty for customers. As such the business is now working on further potential use cases for AI and machine learning using [Vertex AI](https://cloud.google.com/vertex-ai) (<https://cloud.google.com/vertex-ai>).

For example, Simply Energy is planning to use machine learning to help one of its customers select a carbon emissions-reducing biomass fuel source for its electricity and steam-based fuel boiler systems. "With machine learning, we would be able to amalgamate at least 30 different scenarios involving, price, weather, grid-level forecasts and more into one model for comparison," says Pericherla.

The business is now fully empowered to help customers achieve their carbon emission reduction targets. "With Google Cloud technologies and models, we can capture, store, analyze and manipulate data to help our customers move to net zero and keep New Zealand on the path to long-term environmental sustainability," she says.

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